

PRAKTIKUM SISTEM OPERASI

MODUL 5

Eksplorasi Proses Xinu



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

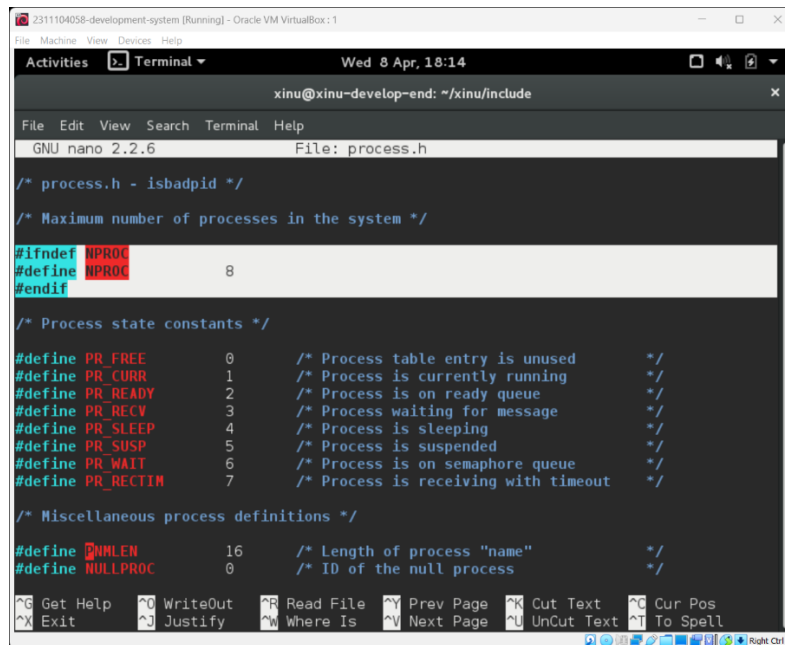
UNIVERSITAS TELKOM

2026

JURNAL

1. Pertanyaan

a. Berapa banyaknya maksimum proses yang ada pada Xinu?



```
2311104058-development-system [Running] - Oracle VM VirtualBox: 1
File Machine View Devices Help
Activities Terminal Wed 8 Apr, 18:14
xinu@xinu-develop-end: ~/xinu/include
File Edit View Search Terminal Help
GNU nano 2.2.6 File: process.h

/* process.h - isbadpid */

/* Maximum number of processes in the system */
#ifdef NPROC
#define NPROC 8
#endif

/* Process state constants */

#define PR_FREE 0 /* Process table entry is unused */
#define PR_CURR 1 /* Process is currently running */
#define PR_READY 2 /* Process is on ready queue */
#define PR_RECV 3 /* Process waiting for message */
#define PR_SLEEP 4 /* Process is sleeping */
#define PR_SUSP 5 /* Process is suspended */
#define PR_WAIT 6 /* Process is on semaphore queue */
#define PR_RECTIM 7 /* Process is receiving with timeout */

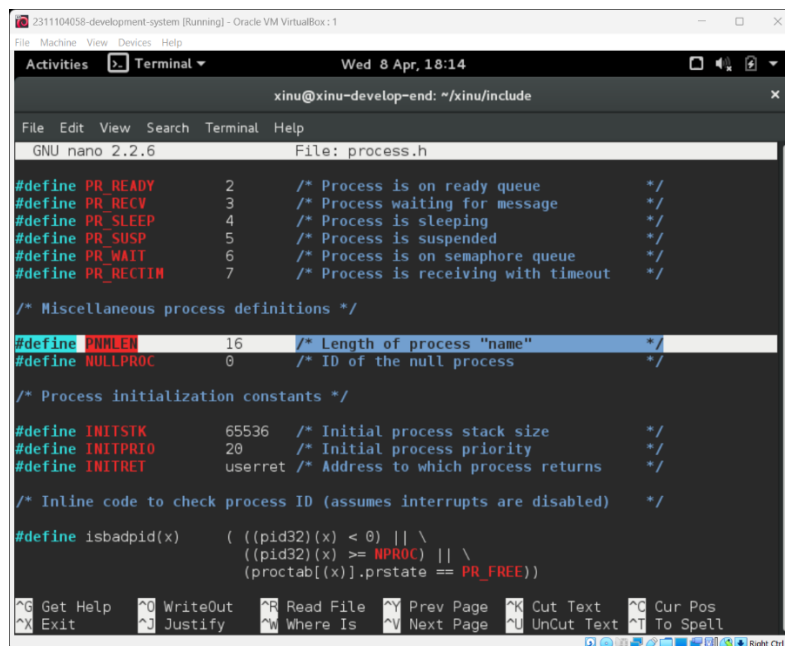
/* Miscellaneous process definitions */

#define NMLEN 16 /* Length of process "name" */
#define NULLPROC 0 /* ID of the null process */

^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^X Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell
```

Maksimum proses pada Xinu adalah 8 proses

b. Berapa maksimal panjang nama suatu proses pada Xinu?



```
2311104058-development-system [Running] - Oracle VM VirtualBox: 1
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Activities Terminal Wed 8 Apr, 18:14
xinu@xinu-develop-end: ~/xinu/include
File Edit View Search Terminal Help
GNU nano 2.2.6 File: process.h

#define PR_READY 2 /* Process is on ready queue */
#define PR_RECV 3 /* Process waiting for message */
#define PR_SLEEP 4 /* Process is sleeping */
#define PR_SUSP 5 /* Process is suspended */
#define PR_WAIT 6 /* Process is on semaphore queue */
#define PR_RECTIM 7 /* Process is receiving with timeout */

/* Miscellaneous process definitions */

#define NMLEN 16 /* Length of process "name" */
#define NULLPROC 0 /* ID of the null process */

/* Process initialization constants */

#define INITSTK 65536 /* Initial process stack size */
#define INITPRIO 20 /* Initial process priority */
#define INITRET userret /* Address to which process returns */

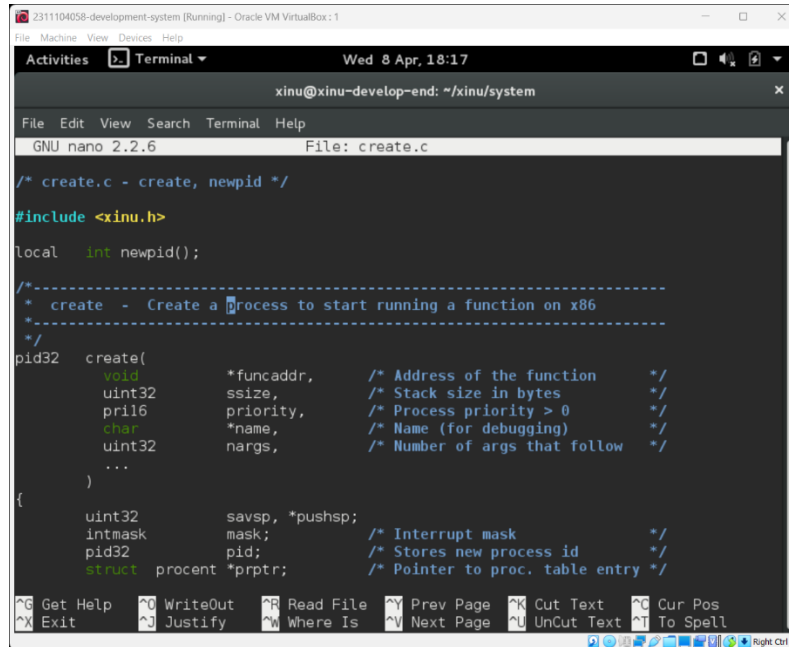
/* Inline code to check process ID (assumes interrupts are disabled) */

#define isbadpid(x) ((pid32)(x) < 0) || \
((pid32)(x) >= NPROC) || \
(proctab[(x)].prstate == PR_FREE)

^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^X Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell
```

Maksimal panjang nama proses adalah 16 karakter

- c. Berapa nilai prioritas awal pada saat proses dibuat?



```
GNU nano 2.2.6 File: create.c

/* create.c - create, newpid */

#include <xinu.h>

local int newpid();

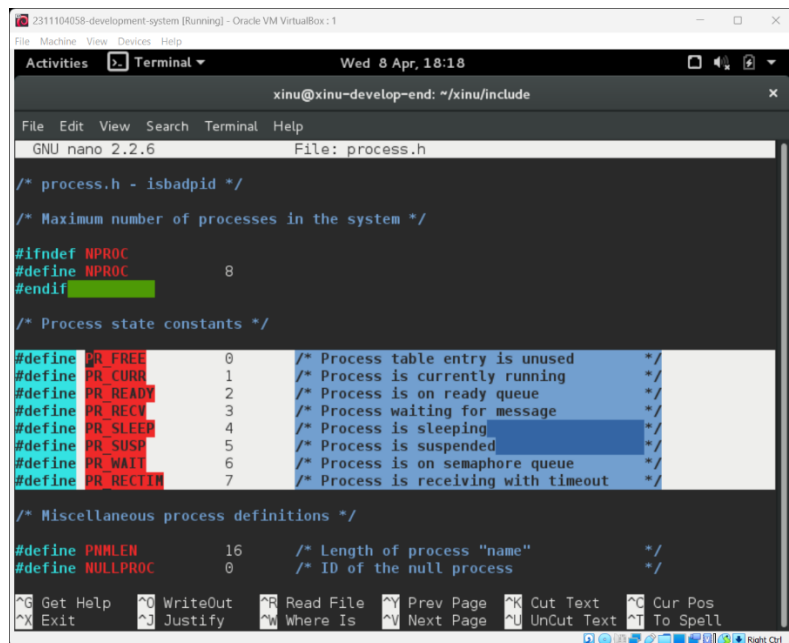
/*-----
 * create - Create a process to start running a function on x86
 *-----
 */

pid32 create(
    void          *funcaddr, /* Address of the function */
    uint32        ssize,     /* Stack size in bytes */
    priority      priority,  /* Process priority > 0 */
    char          *name,     /* Name (for debugging) */
    uint32        nargs,     /* Number of args that follow */
    ...
)
{
    uint32        savsp, *pushsp;
    intmask       mask;      /* Interrupt mask */
    pid32         pid;       /* Stores new process id */
    struct procnt *prptr;    /* Pointer to proc. table entry */

    ...
}
```

Nilai prioritas awal ditentukan oleh parameter priority saat proses dibuat

- d. Ada berapa total state pada Xinu? Sebutkan!



```
GNU nano 2.2.6 File: process.h

/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC 8
#endif

/* Process state constants */

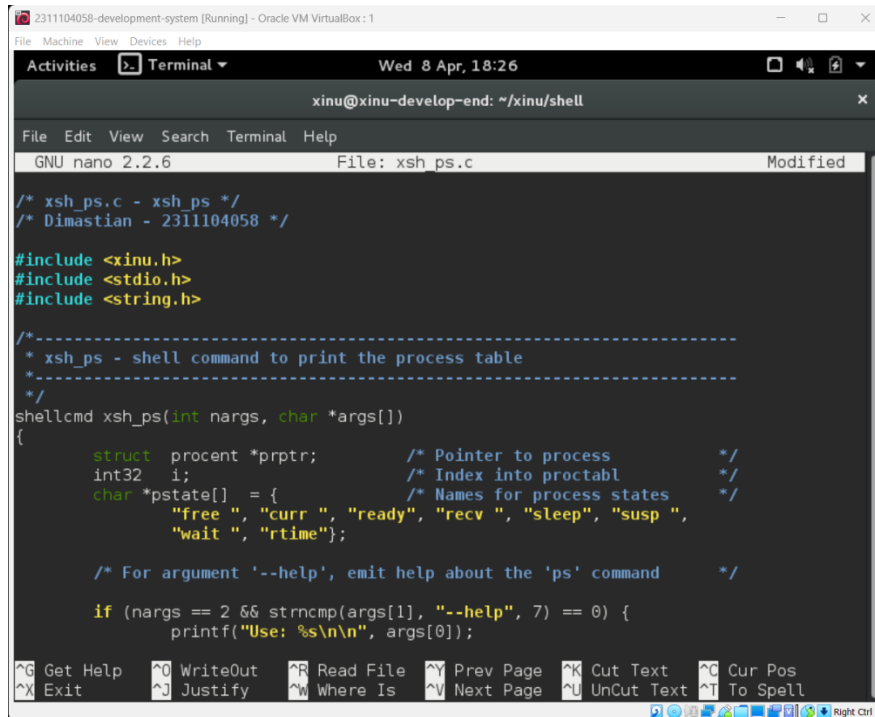
#define PR_FREE 0 /* Process table entry is unused */
#define PR_CURR 1 /* Process is currently running */
#define PR_READY 2 /* Process is on ready queue */
#define PR_RECV 3 /* Process waiting for message */
#define PR_SLEEP 4 /* Process is sleeping */
#define PR_SUSP 5 /* Process is suspended */
#define PR_WAIT 6 /* Process is on semaphore queue */
#define PR_RECVIM 7 /* Process is receiving with timeout */

/* Miscellaneous process definitions */

#define PNMLEN 16 /* Length of process "name" */
#define NULLPROC 0 /* ID of the null process */
```

Terdapat 8 state proses pada Xinu yaitu PR_FREE, PR_CURR, PR_READY, PR_RECV, PR_SLEEP, PR_SUSP, PR_WAIT, PR_RECTIM

2. Perintah ps adalah perintah untuk menampilkan statistik process yang berjalan. Source code dari ps tersimpan pada file xsh_ps.c. Carilah file tersebut dan beri komentar pada 20 baris terakhir di source code tersebut!



```
2311104058-development-system [Running] - Oracle VM VirtualBox: 1
File Machine View Devices Help
Activities Terminal Wed 8 Apr, 18:26
xinu@xinu-develop-end: ~/xinu/shell
File Edit View Search Terminal Help
GNU nano 2.2.6 File: xsh_ps.c Modified

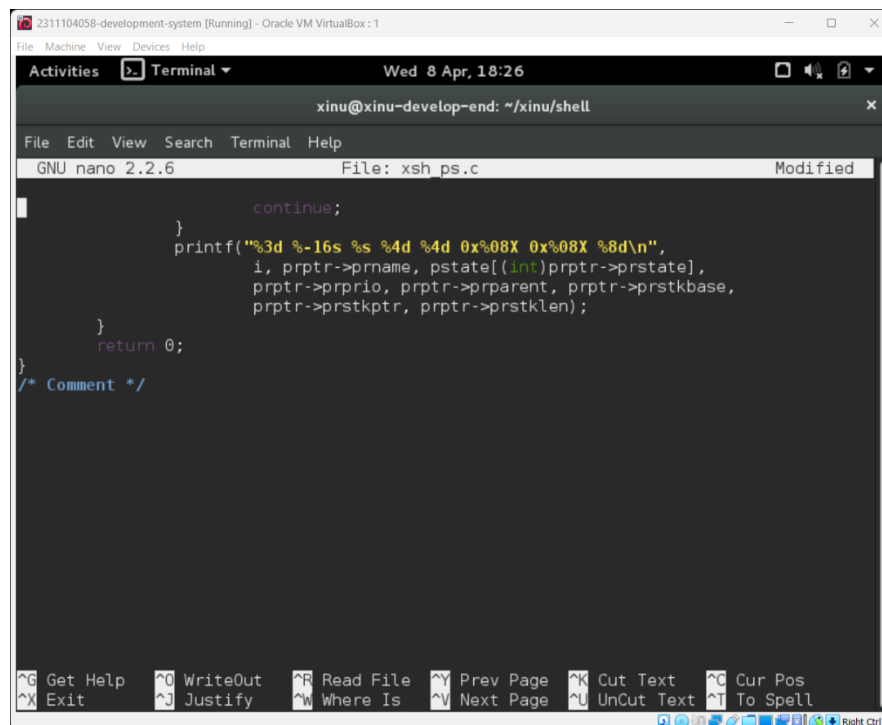
/* xsh_ps.c - xsh_ps */
/* Dimastian - 2311104058 */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_ps - shell command to print the process table
 *-----
 */
shellcmd xsh_ps(int nargs, char *args[])
{
    struct procent *prptr;      /* Pointer to process      */
    int32 i;                   /* Index into proctabl     */
    char *pstate[] = {         /* Names for process states */
        "free ", "curr ", "ready", "recv ", "sleep", "susp ",
        "wait ", "rtime"};

    /* For argument '--help', emit help about the 'ps' command */

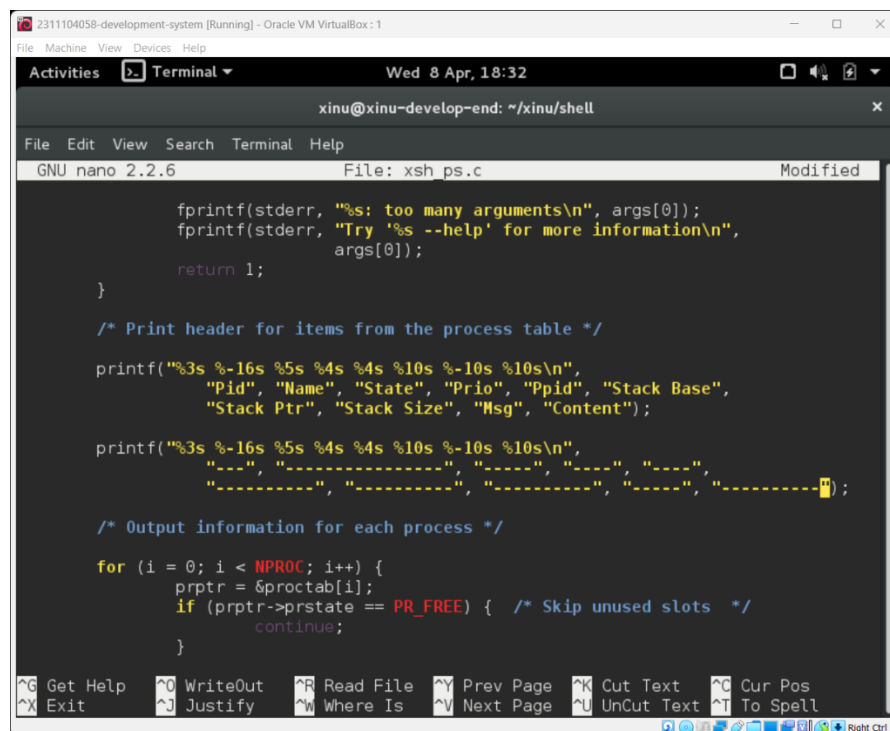
    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n\n", args[0]);
    }
}
```



The screenshot shows a terminal window titled "xinu@xinu-develop-end: ~/xinu/shell" with the GNU nano 2.2.6 editor open to the file "xsh_ps.c". The code visible is a function that prints process information. The terminal window has a menu bar with "File", "Machine", "View", "Devices", and "Help". The status bar at the bottom shows various keyboard shortcuts like ^G Get Help, ^X Exit, ^O WriteOut, ^J Justify, ^R Read File, ^W Where Is, ^Y Prev Page, ^V Next Page, ^K Cut Text, ^U UnCut Text, ^C Cur Pos, and ^T To Spell.

```
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
        i, prptr->prname, pstate[(int)prptr->prstate],
        prptr->prprio, prptr->prparent, prptr->prstkbase,
        prptr->prstkptr, prptr->prstklen);
}
return 0;
/* Comment */
```

3. Ubahlah perintah ps (source code: xsh_ps.c) pada Xinu sehingga menampilkan informasi tambahan berupa kolom yang berisi total message yang ada pada proses
- Ubahan



The screenshot shows the same terminal window as before, but the code in "xsh_ps.c" has been updated. It now includes error handling for too many arguments, a header for the process table output, and a loop to iterate through the process table, skipping unused slots. The terminal window title is "xinu@xinu-develop-end: ~/xinu/shell" and the status bar shows the same keyboard shortcuts.

```
fprintf(stderr, "%s: too many arguments\n", args[0]);
fprintf(stderr, "Try '%s --help' for more information\n",
        args[0]);
return 1;
}

/* Print header for items from the process table */
printf("%3s %-16s %5s %4s %10s %-10s %10s\n",
    "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
    "Stack Ptr", "Stack Size", "Msg", "Content");

printf("%3s %-16s %5s %4s %10s %-10s %10s\n",
    "----", "-----", "-----", "-----", "-----", "-----",
    "-----", "-----", "-----", "-----");

/* Output information for each process */
for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
}
```

```
GNU nano 2.2.6 File: xsh ps.c Modified

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
      "----", "-----", "-----", "-----", "-----", "-----", "-----");

/* Output information for each process */

for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
        i, prptr->prname, pstate[(int)prptr->prstate],
        prptr->prprio, prptr->prparent, prptr->prstkbase,
        prptr->prmsg);
    return 0;
}
/* Comment */
```

Cek Hasil

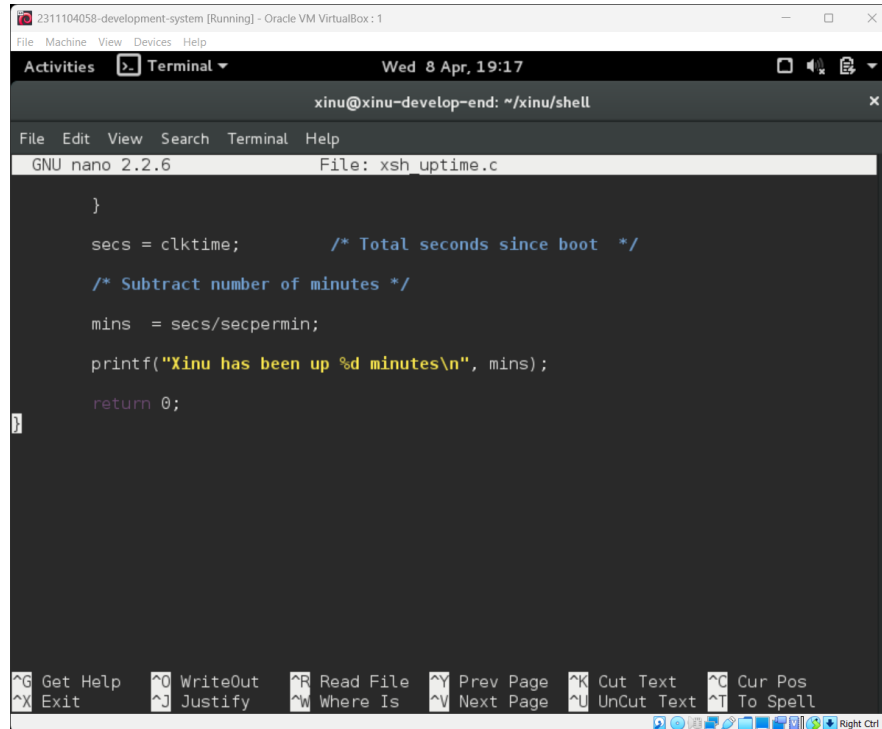
```
-----
XINU
-----

Welcome to Xinu!

xsh $ ps
command 06(xinu) not found
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size
-----
0 prnull          ready  0    0  0x005FDFFC 0x00FFFE4  8192
1 ipout           wait   500  0  0x005FBFFC 0x005FBF30 8192
2 netin           wait   500  0  0x005F9FFC 0x005F9ED0 8192
4 Main process    recv   20   3  0x005E7FFC 0x005E7F34 65536
5 shell           recv   50   4  0x005D7FFC 0x005D79AC 8192
6 ps              curr   20   5  0x005F7FFC 0x005F7FC4 8192
xsh $
```

- Ubahlah perintah uptime pada Xinu sehingga menampilkan lamanya Xinu sejak booting hanya dalam satuan menit.

Ubahan



The screenshot shows a terminal window titled "xinu@xinu-develop-end: ~/xinu/shell". The terminal is running the GNU nano 2.2.6 text editor, editing the file "xsh uptime.c". The code in the file is as follows:

```
}

secs = clktime;      /* Total seconds since boot */

/* Subtract number of minutes */

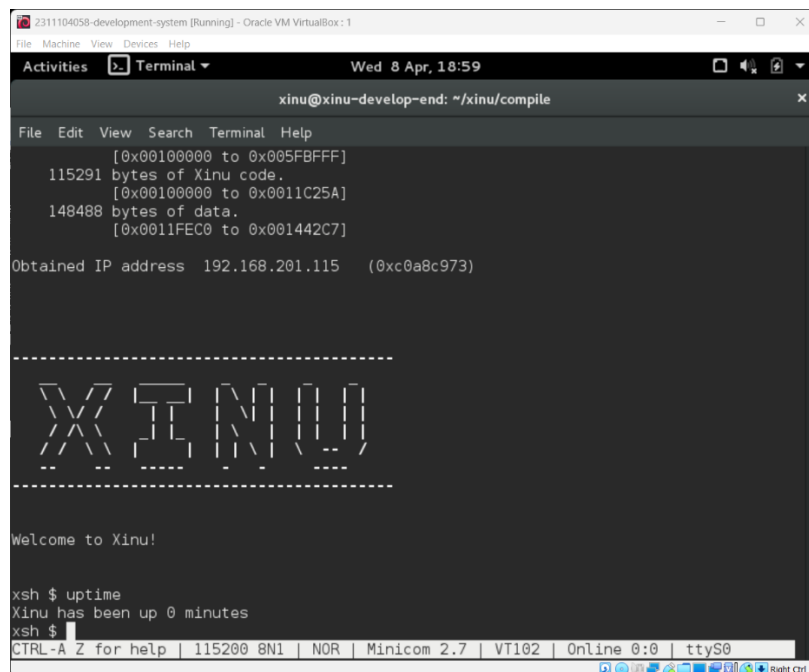
mins = secs/secpermin;

printf("Xinu has been up %d minutes\n", mins);

return 0;
```

The terminal window also shows a status bar at the bottom with various keyboard shortcuts and the current cursor position.

Cek Hasil



The screenshot shows a terminal window titled "xinu@xinu-develop-end: ~/xinu/compile". The terminal displays the output of the Xinu boot process, including the memory layout and the IP address obtained. The output is as follows:

```
[0x00100000 to 0x005FBFFF]
115291 bytes of Xinu code.
[0x00100000 to 0x0011C25A]
148488 bytes of data.
[0x0011FEC0 to 0x001442C7]

Obtained IP address 192.168.201.115 (0xc0a8c973)

-----
XINU
-----

Welcome to Xinu!

xsh $ uptime
Xinu has been up 0 minutes
xsh $
```

The terminal window also shows a status bar at the bottom with various keyboard shortcuts and the current cursor position.